

4,135 ft

Level 2 certification flight apogee

Mach 1.2

peak speed on my second rocket

4

documented flight tests on 2 vehicles

3.97

GPA · President's Honor List

Two rockets, four flights, summer 2026

#	DATE	VEHICLE	APOGEE	MAX VELOCITY	RECOVERY	OUTCOME
L2 — first attempt	Jul 18, 2026	Rocket 1	4,224 ft	659 ft/s	Drogue at apogee · main charge fired at 694 ft · canopy never inflated	Failure Level 2 attempt failed · fins damaged · recovery train redesigned
L2 — certification flight	Aug 8, 2026	Rocket 1 (rebuilt)	4,135 ft	658 ft/s	Drogue at apogee · main at 1,498 ft · 19 ft/s descent	Nominal NAR Level 2 certification on a J250W (~704 N·s) · 6.8° tilt at burnout
Subsonic shakedown	Aug 8, 2026	Rocket 2	2,833 ft	481 ft/s	Drogue at apogee · main at 1,499 ft · 18 ft/s descent	Nominal Nominal shakedown · vehicle cleared for the supersonic attempt
Supersonic attempt	Aug 15, 2026	Rocket 2	Breakup at ≈ 1,800 ft	1,367 ft/s (Mach 1.2)	Avionics bay recovered intact with the full flight record	Failure Structural failure at t = 2.20 s · fin failure identified · redesign specified

Supersonic flight test and failure reconstruction

Scratch-built Blue Tube airframe, 3.1 in OD / 3.0 in ID, 82 in long · 54 mm fiberglass motor mount · AeroTech L1000W, ~2,714 N·s · fiberglass ogive nosecone · fiberglass clipped-delta fins, 5.00 in root chord, 0.25 in thick · dual-deploy recovery · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

My first Level 2 attempt failed right in front of me: the drogue came out, the main never did. I found the cause in the recovery train — I had accounted for the length of the shock cord but not for the shroud lines — restaged the main parachute directly behind the nosecone, and flew again.

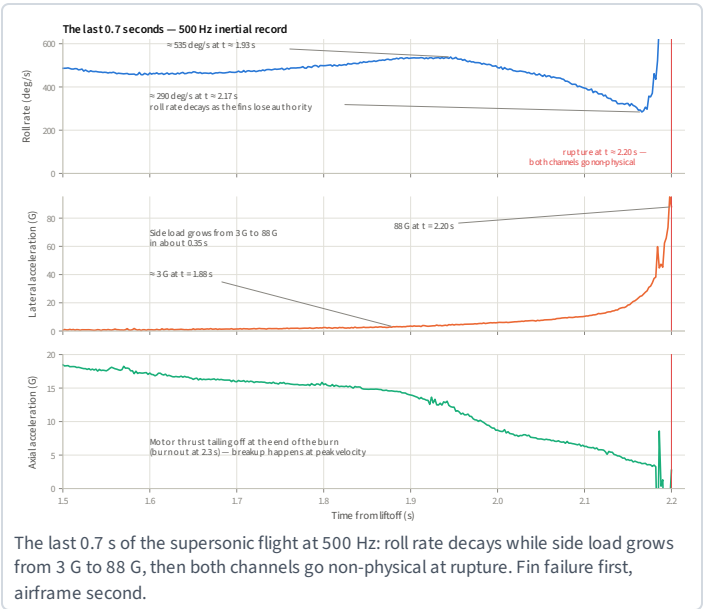
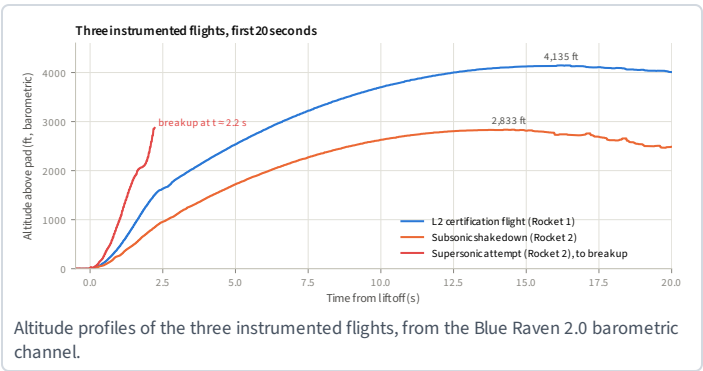
The certification flight earned my Level 2: a J250 motor to 4,135 feet, within 6.8 degrees of vertical at burnout, main deployment at 1,498 feet and a 19 ft/s descent, on a 3D-printed electronics bay I designed that survived a 59.6 G touchdown.

Dual-deploy high power rocket — NAR Level 2 certification

3.1 in OD / 3.0 in ID Blue Tube airframe · 54 mm motor mount · AeroTech J250W, ~704 N·s · 3D-printed PETG ogive nosecone · birch plywood clipped-delta fins, 4.30 in root chord · dual-deploy recovery, 48 in main · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

I flew the second rocket on a subsonic shakedown to 2,833 feet first, then on the supersonic attempt, where it reached 1,367 ft/s (about Mach 1.2) at 2.14 s and broke apart at 2.20 s. The avionics bay came down intact with the complete 500 Hz flight record.

The data showed side loading rising from 3 G to 88 G in about a third of a second while the roll rate decayed from about 535 to 290 deg/s, and all four fins were missing from the wreckage: the fins failed first and the airframe opened afterward. I also caught that the altimeter's 7,131 ft summary altitude was an artifact of the exposed static port and threw it out. Redesign specified: thicker, stiffer fins with a stronger root attachment and a composite airframe.



Also: Co-lead, Launch Operations, Project EXO Bronco (Bronco Launch two-stage rocket aimed at the Karman line, sub-team of 4) · Sub-team lead, Small Rocket Competition (4-person team, 2,800 ft target, onboard temperature and altitude logger I programmed) · SR-30 turbojet test operator to 70,000 RPM · Design Liaison, PIV design team. Full write-ups: [edwincallesmolina.github.io](https://github.com/edwincallesmolina).